

Continuity and Sample-Rate Validation of the `rp_rin_stream` Acquisition Pipeline

Red Pitaya STEMLab 125-14 Z7020-LN — analysis commit 97c8d6c

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Abstract. We validate the `rp_rin_stream` dual-channel ADC streaming pipeline on real hardware by driving both Red Pitaya inputs with a 100 kHz sine from an AWG and capturing 60 s at decimation 128 ($F_s = 125,000,000/128 = 976,562.5$ Sa/s/ch). Four orthogonal signal-level tests all pass for both channels with an injection-validated detection floor of ≈ 1 lost sample at any pack boundary; the capture is therefore *continuous*, contains the expected number of cycles, and has a constant sample rate equal to 125 MHz/128 within the Red Pitaya quartz tolerance ($\sim \pm 50$ ppm).

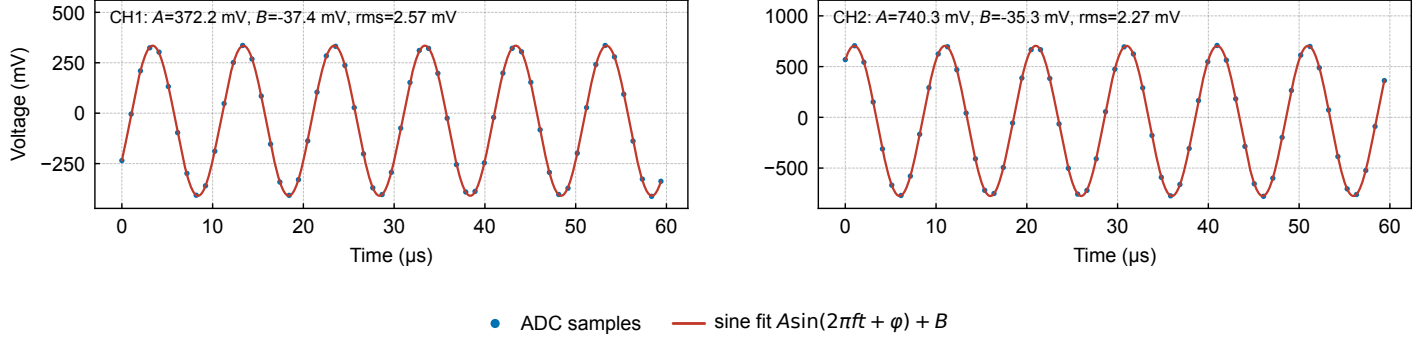


Figure 1: Six cycles of CH1+CH2 (markers, anchored at sample $224 \times 131,072$) overlaid with the LSQ fit $A \sin(2\pi ft + \phi) + B$ at $f_{\text{actual}} = 100,000.4435$ Hz (1 s window; 60 s FFT $\Rightarrow 100,000.5331$ Hz). Residual RMS 2.3–2.6 mV is dominated by AWG distortion + broadband noise, $\sim 100\times$ above the $\sim 20 \mu\text{V}$ ADC quant+thermal floor at $d=128$ with CIC averaging.

Test	CH1	CH2
T1 cycle count	6,000,032 (+0.0125 vs. exp.)	6,000,031 (−0.9875)
T2 coherence ratio	0.9433	0.9559
T3 max —phase jump—	0.0425° (rms 0.0123°)	0.0230° (rms 0.0086°)
T3 SNR vs. $K=1$ ref (36.86°)	$\approx 868\times$ peak / $\approx 3000\times$ rms	$\approx 1602\times$ peak / $\approx 4300\times$ rms
T4 boundary 4σ outliers	0 of 447	0 of 447
Verdict	PASS	PASS

Table 1: $f_{\text{actual}} = 100,000.5331$ Hz from the full-record FFT ($\Delta = +5.33$ ppm vs. nominal 100 kHz). T3 self-test: $K=1 \rightarrow -36.87^\circ$, $K=5 \rightarrow +175.67^\circ$, $K=10 \rightarrow -8.65^\circ$, $K=39 \rightarrow +2.29^\circ$ (under T3 but caught by T1: $K \cdot f_{\text{actual}}/F_s = 3.994$ cycles $\gg$ T1's ≤ 1), $K=100 \rightarrow -86.44^\circ$, $K=1000 \rightarrow -144.21^\circ$. T1+T3 jointly cover all isolated $K \geq 1$ except integer-cycle K within ± 0.01 cyc (T2 is a loose backstop).

1 Setup and Launcher Verdict

A single AWG drove ADC inputs CH1 (≈ 0.38 V peak) and CH2 (≈ 0.75 V). Capture: `./scripts/run_acquire.sh -d 128 -t 60 (LV  $\pm 1$  V, BIT16 raw)`. Launcher passed -1 58,593,750 (= $60F_s$) to the official ARM C++ `rpasa_client`, which streamed one `waveform.bin` into `/tmp` on the board and `scp'd` it back. `summary.json` reports `state=finished`, `samples_received_per_channel=58,593,750` (matches the requested limit exactly), all four loss counters = 0, `pos_sequence_gaps_total=0`, `file_pack_count=448` (= 447 full + 1 truncated tail). The receiver count $448 \times 131,072 = 58,720,256$ is reconciled with the on-disk 58,593,750 by a single in-pack truncation of the final pack ($\Delta = 126,506 < 131,072$, less than one full pack), as designed by -1.

2 Methodology

Four orthogonal tests in `tests/verify_100kHz_capture.py`, jointly flagging any pack-boundary or global-rate discontinuity the launcher might miss (sub-pack discontinuities are architecturally impossible: the FPGA writes contiguous decimated samples to a DMA buffer that `rpasa_client` then slices into packs). **T1.** Cycle accounting — pos.-going zero-crossings vs. $f_{\text{actual}} \times \text{duration}$; threshold ≤ 1 cycle. **T2.** Full-record coherence — coherent inner product at f_{actual} over 60 s, ratioed to peak-to-peak/2 (rules out gross corruption; ratio > 0.85). **T3.** Per-pack-boundary phase jump — phase difference between left $[b-W, b]$ and right $[b, b+W]$ windows ($W=4096$) referenced to the same instant; a hidden K -sample gap injects $\text{wrap}(2\pi f_{\text{actual}} K/F_s)$ at full amplitude (e.g. $K=1 \Rightarrow 36.86^\circ$); threshold 5° . **T4.** Boundary vs. interior 1st-difference — distribution of $|x[b] - x[b-1]|$ at boundaries vs. random interior positions; outlier count at $4\sigma_{\text{interior}}$ (frequency-domain-independent). A *sensitivity self-test* injects synthetic K -sample gaps to verify T3 registers the analytic excursion.

3 Sample-Rate Stability and Scope

Red Pitaya FPGA `osc_decimator.v` is a synchronous integer counter on the on-board 125 MHz quartz; for $d \geq 16$ each output sample is the non-overlapping CIC average of d input samples (1.024 μs window at $d=128$), preserving the $1/F_s$ spacing relied on by all four tests and giving built-in anti-aliasing below $F_s/2$. No software resampling anywhere. **Scope:** validated for 60 s at $d=128$ with a clean coherent in-band tone; not for signals near $F_s/2$, sub-LSB amplitudes, runs < 10 packs, correlated channel-pair errors, or in-segment DC/gain drift. Audited by Anthropic Opus 4.7 and OpenAI Codex GPT-5.5 *xhigh*.